

Supplementary Material

Metamorphic Debugging of Learned Components: Interaction Requirements for Contrastive VLMs

This document gives the specification and repair settings, the color-binding coverage argument, and the supporting results that the paper cites. Learned results use seeds 42, 43, and 44; $\pm$ denotes their sample standard deviation. Confidence intervals are paired 95% seed/source bootstrap intervals, with all edits and captions of a source kept together: 5,000 draws for typography and color binding, and 4,000 for backdoors. Each draw resamples the three seeds and source clusters with replacement, sharing both draws across methods. Color-binding clusters join scenes that share a source photograph. Behavioral values are percentages unless stated otherwise. Bold marks the best displayed mean for each metric, including ties.

S1 Specification and Repair Settings

S1.1 A worked metamorphic specification

The recognition requirement is to identify the depicted object despite irrelevant writing. Its metamorphic specification fixes the source object and note geometry, varies the writing and the class description, and targets zero change in the object–distractor margin. The developer supplies this relationship and the valid edits; the implementation constructs the related executions and evaluates their scores.

The running example is one instance of the specification: COCO image 284084 (annotation 1339759), the fixed-note renderer, and the classes car and handbag represented by the pooled typography templates below. Positive clean and blank-note margins are retention checks. For each rendered state r and class c , the audit computes $S_{r,c} = s(\text{render}(\text{source}, r), \text{caption}(c))$, the margins $m_r = S_{r,\text{car}} - S_{r,\text{handbag}}$, $D_{\text{blank},d} = m_{\text{word}} - m_{\text{blank}}$, and the failure flag $(m_{\text{blank}} > 0) \wedge (m_{\text{word}} \leq 0)$, and records them with the source and checkpoint identities. For the typographic source in Figure 1, $D_{\text{blank},d}$ is about $-.1176$ for the frozen model, $-.0778$ after ranking, and $-.0475$ after MIRROR; only MIRROR corrects the word decision. The developer supplies the factors, targets, edits, and retention checks; the software computes the score differences and the gradient calibration. Typography instantiates the relation for all six pairs of four note states against each of two object–distractor caption contrasts: 12 interaction measurements per source. With 1,024 primary held-out sources this yields 12,288 measurements per model; recognition of each misleading note against its named alternative gives 2,048 pairwise decisions. Figure 1 displays one of those failing instances before repair.

S1.2 Rendering and optimization

Every configuration uses seeds 42, 43, and 44 with coefficients fixed across seeds.

Color binding. COCO masks define two object cutouts on a 256×256 gray canvas. Cutouts fit a 112-pixel maximum dimension with fixed margins and vertical centering. Masked pixels combine 25% original RGB with 75% luminance-modulated tint; unmasked pixels are unchanged. The five training pairs are bus/airplane, car/boat, cat/couch, dog/sports ball, and person/bicycle. The 400-source primary test uses new sources for the four pairs excluding dog/sports ball. Captions use “a c_1 A and a c_2 B,” “a c_1 A next to a c_2 B,” and “the A is c_1 and the B is c_2 .” Training cycles individual and pooled templates equally, with both physical layouts and balanced canonical/reversed noun

order (18 epochs each). Reversing noun order preserves the semantic object slots. The primary test pools red/blue and green/yellow, both noun orders, and both layouts within each source. The training configuration is selected on 888 source-disjoint validation pairs.

Calibration uses all 70 red/blue examples in the fixed, source-hashed Visual Genome split. Let m be the correct-minus-opposite color margin with pooled templates: “a {color} {noun},” “a photo of a {color} {noun},” and “the {noun} is {color}.” Each frozen backbone uses $u = \text{median } |m|$ (.04815875 for LAION L/14). The LAION B/32 reference fixes shared bounds $K = Q_{.25}(|m|)/u = .5175$ and $\tau = \beta = \max\{Q_{.75}(|m^{(1)} - m^{(2)}|)/u, .05\} = .08449$, using separate first/second-template margins $m^{(1)}, m^{(2)}$. Bounds are reused in each backbone’s frozen unit. Write D for a binding contrast and C for a cross-effect. We write $e = D_{\text{assign}}/2$ for half the assignment interaction and b for the shared caption preference; both are defined in Supplement S2. The main paper reports the full assignment interaction D_{assign} . With frozen values denoted by superscript 0, targets are $D^* = \max(D^0, K)$, $C^* = \min(|C^0|, \tau)$, $e^* = \max(e^0, K - \tau)$, and $b^* = \min(|b^0|, \beta)$. Losses penalize positive shortfalls for D, e and positive excesses for $|C|, |b|$, with slack 10^{-5} . Each averages its clauses, takes the maximum over paired layouts, then averages blocks. Their weights are $w_D, 4w_C, w_e, 4w_b$, plus four times the retention loss. Gradient normalization uses eight fixed batches. With component mean gradient norms g_j and their positive median g_* , $w_j = \text{clip}(g_*/g_j, .1, 10)$, with zero norms assigned weight 10. The reference median includes the four targets, four margin retention terms below, and the exchange-margin shortfall $[\max(m^0, K - \tau - \beta) - m - 10^{-5}]_+$.

For LAION L/14, the weights are

$$(w_D, w_C, w_e, w_b) = (1, .69365, 5.54663, .58075).$$

For a caption, binding, or color-free object margin z , retention uses $\mathbf{1}[z^0 > 0] [z^0 - z - 10^{-5}]_+$; response retention uses $[e^0 - e - 10^{-5}]_+$. Weights in caption/binding/response/object order are (.43590, 1.16543, 8.07771, .16248). Unit-weight task/natural-caption drift uses squared Euclidean distance between normalized adapted and frozen embeddings. All retention terms use the same clause/layout reduction. Both ranking and MIRROR additionally retain single-word color margins z with $[\max(z^0, m_0) - z - 10^{-5}]_+$, averaged over clauses, maximized over layouts, and averaged over blocks. The floor $m_0 = .18091843$ is one quarter of the median positive frozen training margin in calibrated units. Its gradient-normalized weight is .45021634, multiplied by the shared retention priority four.

AdamW uses learning rate $2 \cdot 10^{-4}$, weight decay .01, gradient clipping at one, batch 24 paired blocks, and 36 epochs. The map has rank/alpha 64 and dropout .05. The main Ranking baseline replaces the interaction and preference targets with CE(100s) and retains MIRROR’s complete retention losses, including binding/response retention. The score scale 100 is fixed across these color binding comparisons. All color binding comparisons use the same 75% tint, balanced noun orders, source sets, and evaluation averaging. OpenAI L/14 and SigLIP B/16 retain their fixed backbone-specific target and retention weights. The single-color floor is calibrated on training inputs only: (m_0, w_0) is (0.35457206, 0.37220872) for OpenAI L/14 and (0.20701225, 0.34407193) for SigLIP B/16. These backbone replications perform no additional hyperparameter search.

Typography. Annotated boxes have minimum side 64 pixels and at least 8% image area. A 10%-padded crop is padded to 224×224 . The white note spans (11, 174)–(212, 212); black DejaVu Sans Bold text is sized to fit the longest word for that source. Transfer uses DejaVu Serif or moves the note to (11, 11)–(212, 49). Text features average three templates: “a photo of a {class},” “a {class},” and “an image of a {class}.” The shared token follows the start token and is initialized from $\mathcal{N}(0, .02^2)$; it has 512 parameters on B/32 and 768 on L/14. Both repairs use CE(100s), three times clean/blank

frozen-reference KL, a squared object-margin floor at .045 with weight 1.2, and squared clean/blank mean-margin retention divided by .1². MIRROR adds $\kappa \mathbb{E}_{d,r < r'} (m_{r,d} - m_{r',d})^2 / .1^2$ over the four note states. The coefficients are $\kappa = 6.07975$ for OpenAI B/32, 6.06078 for OpenAI L/14, and 4.22886 for LAION B/32. SGD uses learning rate .002 with cosine decay to .00005, batch 32, and no momentum or weight decay. Class-balanced sampling gives 2,752 updates. The 38 held-out classes and evaluation source IDs are excluded from training.

Backdoor. Clean and edited scores use 87 training captions. For true class a define $m_{r,d} = s(x_r, t_a) - s(x_r, t_d)$. Both repairs minimize $\text{CE}(100s) + 3\text{KL}(p_{\text{clean}}^0 \| p_{\text{clean}}) + \mathbb{E}_d \left[m_{0,d}^0 - m_{0,d} \right]_+^2 / .1^2$. MIRROR adds $\kappa \mathbb{E}_{r > 0, d} (m_{r,d} - m_{0,d})^2 / .1^2$. The coefficients are $\kappa = 10.17165$ (stripes), 10.54683 (triangles), and 12.35752 (text). AdamW uses learning rate 10^{-5} , cosine decay to 10^{-6} , and weight decay .001. Batches contain one banana and 31 class-balanced draws. The schedule has 1,536 updates. Clean-only uses four clean views per batch with the same budget. Only the last two visual blocks change. Evaluation uses PAR’s renderer: a 16×16 striped patch at two source-seeded locations, or a blend of strength .15 for triangles and .5 for text. The control uses the source image’s channel-wise mean color on the same support and at the same opacity. ImageNetV2 recognition and trigger interactions use the first triggered view; processing retests apply JPEG qualities 90 and 70 or resize through 168 and 112 pixels, then restore 224×224 . Scores pool the common 80-template ImageNet caption ensemble. Trigger-interaction means exclude the true class, averaging the remaining 999 alternatives per source.

S2 Color-Binding Coverage and Decision Adequacy

For two opposite assignments, let t_0 correctly describe x_0 and t_1 correctly describe x_1 . Their correct-caption margins are $m_0 = s(x_0, t_0) - s(x_0, t_1)$ and $m_1 = s(x_1, t_1) - s(x_1, t_0)$. The assignment interaction is the same mixed finite difference used throughout the paper:

$$D_{\text{assign}} = (s(x_1, t_1) - s(x_1, t_0)) - (s(x_0, t_1) - s(x_0, t_0)) = m_0 + m_1.$$

Define $e = D_{\text{assign}}/2$ and $b = (m_0 - m_1)/2$. Then $m_0 = e + b$, $m_1 = e - b$, and both decisions are correct exactly when $e > |b|$, or equivalently $D_{\text{assign}} > |m_0 - m_1|$. A nonpositive assignment interaction cannot make both decisions correct. With a positive interaction, a shared caption preference can still overturn one of them. This first-order effect does not appear in an interaction because caption-only score offsets cancel. The repair therefore has a separate caption-preference target.

Index each image’s colors by $(x_1, x_2) \in \{0, 1\}^2$ and caption colors by $(y_1, y_2) \in \{0, 1\}^2$. Let $D_{ij}(p, q)$ be the interaction between x_i and y_j , conditioning the other image factor on p and the other caption factor on q . For the assignment exchange $01 \leftrightarrow 10$, expansion of the four-score contrasts gives

$$2e = D_{11}(1, 1) - D_{12}(1, 1) - D_{21}(1, 1) + D_{22}(1, 1). \quad (\text{S1})$$

Thus if the two binding terms are at least K , the two absolute cross-effects at most τ , and $|b| \leq \beta$, both correct-caption margins are at least $K - \tau - \beta$. In particular, $K > \tau + \beta$ suffices for both decisions.

Context coverage is necessary. Ordering states as (00, 01, 10, 11), consider

$$S = \begin{pmatrix} .12 & .06 & .06 & 0 \\ .06 & .12 & 0 & .06 \\ .06 & 0 & .12 & .06 \\ 0 & .06 & .06 & .12 \end{pmatrix}, \quad S' = \begin{pmatrix} .12 & .06 & .06 & 0 \\ .06 & .12 & .20 & .06 \\ .06 & .20 & .12 & .06 \\ 0 & .06 & .06 & .12 \end{pmatrix}.$$

A suite measuring binding only at contexts (0, 0), (1, 1) and cross-effects only at (0, 1), (1, 0) gives identical values for S and S' , but the exchange margins change from $+.12$ to $-.08$. Both matrices are cosine-realizable: choose coordinate image vectors, place each score column in a text vector’s first four coordinates, and complete its unit norm in a separate coordinate. This counterexample motivates complete conditional coverage.

S3 Typography: Penalty Strength and Preservation

We vary the coefficient κ of the interaction penalty on OpenAI B/32, keeping the training images, token initialization, source order, retention losses, and 2,752-update budget fixed. Ranking sets this coefficient to zero. Table S1 compares two nonzero coefficients; the larger is the setting used in the main paper. Both use the final checkpoint. Digital tests use 1,024 sources; fixed failures are counted over the 1,547 frozen-model failures. SugarCrepe has 7,511 pairs and RTA100 has 1,000 images. Interactions use raw cosine differences.

Table S1: Typography repair and preservation under two interaction-penalty weights.

Measure	Frozen	Ranking	MIRROR ($\kappa = 1.51994$)	MIRROR ($\kappa = 6.07975$)
Word interaction ↓	0.07704	0.04782 ± 0.00031	0.03950 ± 0.00012	0.02945 ± 0.00057
Object-description accuracy ↑	23.29	59.98 ± 0.67	67.04 ± 0.32	75.44 ± 0.87
Attacked recognition ↑	22.17	55.50 ± 0.47	61.20 ± 0.64	65.89 ± 0.71
Fixed failures ↑	0.00	48.70 ± 0.88	58.03 ± 0.42	69.27 ± 1.25
Clean recognition ↑	79.59	81.02 ± 0.41	81.22 ± 0.25	80.08 ± 0.68
SugarCrepe ↑	76.54	76.07 ± 0.27	75.93 ± 0.43	73.35 ± 1.06
RTA100 recognition ↑	46.70	59.00 ± 0.98	59.33 ± 1.12	54.47 ± 6.04

The larger coefficient reduces mean word interaction from .03950 to .02945 and raises fixed failures from 58.03% to 69.27%. SugarCrepe accuracy falls from 75.93% to 73.35%. Suppression strength therefore controls a measurable repair–preservation trade-off.

S4 Typography Replication Across Backbones

The main typography repair objective is used on all three backbones with the backbone-specific coefficients in Section S1. Each learned method uses three seeds and the same training images, initialization, source order, retention losses, and update budget as its paired comparison. Fresh-source tests use 832 seen-class and 986 unseen-class images. Preservation tests use 1,024 clean sources and all 7,511 SugarCrepe pairs.

Table S2: Typography accuracy gains over ranking on fresh sources (percentage points, paired 95% CIs).

Backbone	Seen classes	Unseen classes
OpenAI B/32	+14.86 [13.14, 16.59]	+8.99 [7.42, 10.56]
OpenAI L/14	+18.09 [15.56, 20.79]	+7.17 [4.06, 10.50]
LAION B/32	+9.84 [7.95, 11.96]	+6.36 [5.17, 7.71]

Table S3: Clean recognition and SugarCrepe preservation on additional typography backbones (%).

Test	Frozen	Ranking	MIRROR
<i>OpenAI L/14</i>			
Clean 70-class recognition	81.93	82.39 ± 0.49	79.26 ± 1.18
Full SugarCrepe	76.83	76.06 ± 0.46	72.18 ± 1.39
<i>LAION B/32</i>			
Clean 70-class recognition	83.50	83.46 ± 1.05	82.71 ± 0.20
Full SugarCrepe	81.20	80.36 ± 1.02	77.44 ± 0.48

S5 Backdoor Target-Class Preservation

This preservation test measures recognition of genuine bananas without a trigger. The test set contains 400 held-out COCO banana crops selected by fixed source hashing. Each model is evaluated among all 1,000 ImageNet classes. Ranking, clean-only fine-tuning, and MIRROR values are means over the three seeds; poisoned and released PAR checkpoints are fixed.

Table S4: Genuine-banana recognition (%) for the three backdoor victims.

Method	Stripes	Triangles	Text
Poisoned	15.25	8.00	13.75
PAR	56.25	58.75	63.25
Clean-only	22.67	12.67	18.67
Ranking	18.83	11.58	16.42
MIRROR	22.00	12.67	19.00